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The Inverse of a Multivector: Beyond the Threshold $p + q = 5$

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Abstract. The algorithm of finding an inverse multivector (MV) numerically and symbolically is of paramount importance in the applied Clifford geometric algebra (GA) $Cl_{p,q}$. The first general MV inversion algorithm was based on matrix representation of MV. The complexity of calculations and the size of the answer in a symbolic form grow exponentially with the dimension $n = p + q$. The breakthrough occurred when Lundholm and then Dadbeh found compact inverse formulas up to dimension $n \leq 5$. The formulas were constructed in a form of Clifford product of the initial MV and carefully chosen grade-negation counterparts. In this report we show that the grade-negated self-product method can be extended beyond the threshold $n = 5$ if, in addition, properly constructed linear combinations of such MV products are used. In particular, we present compact explicit MV inverse formulas for algebras of vector space of dimension $n = 6$ and show that they embrace all lower dimensional cases as well. For readers convenience, we have also given various MV formulas in a form of grade negations when $n \leq 5$.

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1. Introduction

The knowledge of how to find inverse multivectors (MVs) in a Clifford algebra in a symbolic and coordinate-free form is very important both from practical computational and purely theoretical points of view. A universal formula for inverse MV would allow to write down a fast and general algorithm for all occasions rather then to resort to either specific symbolic or numerical cases. The inverse of MV can be used to find explicit solutions of algebraic GA

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equations with all the ensuing consequences and applications. A closely related problem is the normalization of spinors [14,20]. Invertibility also serves as an important criterion on deciding whether homogeneous versors are the blades [3], which are essential in numerous geometric constructions.

The first attempts of inversion of some specific forms of MVs can be traced back to papers [15–17]. However it was not until 2002 when Fletcher [8] suggested some general MV inverse formulas for low dimensional algebras. His algorithm was based on decomposition of MV into matrix basis elements, and, as a consequence, resulted into large, inconvenient and signature dependent formulas the size of which grows exponentially with algebra dimension.

The breakthrough occurred after Lundholm [13] suggested expressions for (determinant) norms when $n \leq 5$; later Dadbeh [6] introduced grade-negation operations which allowed to write down compact and explicit formulas for MV inverse in a coordinate-free form. The heart of the algorithm [6] is the product of the initial MV and carefully chosen grade-negated counterparts that after few iterations eventually yields a scalar. As a matter of fact, the product may be related with the determinant of a matrix representation of general MV, from which the inverse multivector can be easily extracted by simply removing the initial MV which is always positioned in either left-most or right-most side of the product. Using the described method Dadbeh was able to find explicit inverses for a general MV up to dimension $n \leq 5$. It is important to stress that the obtained formulas are determined only by the vector space dimension $n = p + q$ and are independent of the particular signature (p, q). The same formulas were also obtained by other authors using different methods (see, for example, [12,18,19]). When $n \leq 5$, detailed mathematical proofs are given in [12]. If a general multivector A is given in expanded form in some orthogonal basis with symbolic coefficients, then the verification of the algorithm can be easily done by the direct substitution of the symbolic MV into the formula, by explicitly computing the inverse A^{-1}, and by checking that the property $\mathsf{A}\mathsf{A}^{-1} = \mathsf{A}^{-1}\mathsf{A} = 1$ is satisfied. For $n > 4$, however, calculation of explicit inverse in symbolic form is time consuming and results in lengthy expressions in the coefficients of A^{-1} in the given basis. Nonetheless, such calculations, in fact, have status of "computer-assisted proof". Similar formulas for the (determinant) norms of MVs when $n \leq 5$ were also given in [14]. The analysis of the general structure of such formulas was presented in [19]. However, until now all attempts to step beyond the threshold $p + q = 5$ were unsuccessful although there is a need for such formulas in practice.

In this report we show that the grade-negation method can be extended beyond the threshold $n = 5$ if, in addition, properly constructed linear combinations of grade-negated MVs are introduced. In particular, we write down explicit MV inverse formulas for algebras with vector space dimension $n = 6$ having all possible signatures (p, q). We also provide alternative formulas for even MVs in the case $n = 5$. In Sect. 2 the grade-negation method is shortly reviewed and required notation and terminology is introduced. In Sect. 3 the inverse even MVs that follow from higher grade inverse MVs are considered.

Finally, in Sect. 4 a general algorithm to construct inverse MVs for $n = 6$ is briefly discussed and the obtained coordinate-free formulas are presented in the form of tables.

2. Grade-Negated Self-Product and Inverse of a General MV in the Case $n \leq 5$

Following [6] we first introduce a grade-negated self-product that is defined via *grade-r negation operation*. Applied to the multivector A this operation does what it says, i.e., it changes the sign of the grade-r part of A. Such a grade-r negated MV will be denoted as $\mathsf{A}_{\bar{r}}$; the bars over indices designate which grades have opposite signs. Formally the grade-r negated MV can be expressed as $\mathsf{A}_{\bar{r}} = \mathsf{A} - 2\langle \mathsf{A}\rangle_r$, or $\mathsf{A}_{\bar{r},\bar{s}} = \mathsf{A} - 2\langle \mathsf{A}\rangle_r - 2\langle \mathsf{A}\rangle_s$ for a double negation, where $\langle \mathsf{A}\rangle_r$ denotes the grade-r projection of the multivector A. In particular, we have $(\langle \mathsf{A}\rangle_r)_{\bar{r}} = -\langle \mathsf{A}\rangle_r$.

The following properties are evident from the definition of grade-negation operation: $(\mathsf{A}_{\bar{p}})_{\bar{r}} = \mathsf{A}_{\bar{p},\bar{r}} = \mathsf{A}_{\bar{r},\bar{p}}$, $\mathsf{A}_{\bar{r},\bar{r}} = \mathsf{A}$, $(\mathsf{A} + \mathsf{B})_{\bar{r}} = \mathsf{A}_{\bar{r}} + \mathsf{B}_{\bar{r}}$. However, $(\mathsf{AB})_{\bar{r}} \neq \mathsf{A}_{\bar{r}}\mathsf{B}_{\bar{r}}$. If A does not contain grade-r elements then negation returns the same MV, $\mathsf{A}_{\bar{r}} = \mathsf{A}$. The commutator with the grade-negated MV can be expressed as $\mathsf{A}\mathsf{A}_{\bar{r}} - \mathsf{A}_{\bar{r}}\mathsf{A} = 2(\langle \mathsf{A}\rangle_r \mathsf{A} - \mathsf{A}\langle \mathsf{A}\rangle_r)$. The multivector A commutes with the scalar-negated A, i.e., $[\mathsf{A}, \mathsf{A}_{\bar{0}}] = 0$ and, as a consequence, with $\mathsf{A}_{\bar{i},\cdots,\bar{j}}$, where all grades $i \neq 0, j \neq 0, \cdots$ of A (except scalar) are negated.

The standard involutions such as the reversion $\widetilde{\mathsf{A}}$, the grade involution $\widehat{\mathsf{A}}$, and the Clifford conjugation $\widehat{\widetilde{\mathsf{A}}} = \widetilde{\widehat{\mathsf{A}}}$ can be written in terms of negation, respectively, as $\widetilde{\mathsf{A}} = \mathsf{A}_{\bar{2},\bar{3},\bar{6},\bar{7},\bar{10},\bar{11}\ldots}$, $\widehat{\mathsf{A}} = \mathsf{A}_{\bar{1},\bar{3},\bar{5},\bar{7},\bar{9},\bar{11}\ldots}$ and $\widehat{\widetilde{\mathsf{A}}} = \mathsf{A}_{\bar{1},\bar{2},\bar{5},\bar{6},\bar{9},\bar{10}\ldots}$. For finite algebras the index series should break off when the negation index becomes larger than that of the pseudoscalar. It is worth mentioning that the grades $0, 4, 8, 12, \ldots$ are absent in the above list of standard GA involutions. From this follows that the inverse of a general MV cannot be expressed using just standard involutions, or their combinations. As we shall see this observation also is directly related to the MV inversion problem for $n \geq 6$.

By grade-negated self-product we mean the geometric product of a general MV A with any number of its grade-negated counterparts, for example, geometric product $\mathsf{A}\mathsf{A}_{\bar{r}}\mathsf{A}_{\bar{k},\bar{l},\bar{t}}\mathsf{A}_{\bar{s},\bar{t}}\cdots$ is a left grade-negated self-product and $\mathsf{A}_{\bar{r}}\mathsf{A}_{\bar{k},\bar{l},\bar{t}}\mathsf{A}_{\bar{s},\bar{t}}\cdots\mathsf{A}$ is a right grade-negated self-product. These self-products, where the initial MV stands in the left-most or right-most position, will be of special importance in finding the inverse MV A^{-1}.

The explicit inverse formulas which we will construct rely on our ability to get a real scalar using just grade-negated self-products (or, as we shall see, some linear combinations of them for $n > 5$), where the initial MV A must be present in the left-most or right-most position. We shall call the real scalar

$$s_m = \overbrace{\mathsf{A}\mathsf{A}_{\bar{r},\bar{t},\ldots}\mathsf{A}_{\bar{s},\ldots}\cdots}^{m} = \mathsf{A}\, f(\mathsf{A}) = f(\mathsf{A})\,\mathsf{A} \tag{1}$$

obtained in this way the determinant norm[1]. If we succeed in constructing such a scalar s_m then it is easy to see that the inversion formula for A (correspondingly, for $\widetilde{\mathsf{A}}$) can be obtained simply by removing either the extreme left factor A or the extreme right factor A from the product giving the scalar s_m in (1) (correspondingly, from $s'_m = \cdots \widetilde{\mathsf{A}}_{\bar{s},\ldots}\widetilde{\mathsf{A}}_{\bar{r},\bar{t}\ldots}\widetilde{\mathsf{A}})$, and then dividing the result by s_m or $s'_m = \widetilde{\mathsf{A}}\,\widetilde{f}\big(\widetilde{\mathsf{A}}\big) = \widetilde{f}\big(\widetilde{\mathsf{A}}\big)\,\widetilde{\mathsf{A}}$. The inversion formulas for a MV A, thus, become

$$\mathsf{A}^{-1} = \frac{\mathsf{A}_{\bar{r},\bar{t},\ldots}\mathsf{A}_{\bar{s},\ldots}\cdots}{s_m} = \frac{f\big(\mathsf{A}\big)}{s_m}, \quad \widetilde{\mathsf{A}^{-1}} = \frac{\cdots\widetilde{\mathsf{A}}_{\bar{s},\ldots}\widetilde{\mathsf{A}}_{\bar{r},\bar{t}\ldots}}{s'_m} = \frac{\widetilde{f}\big(\widetilde{\mathsf{A}}\big)}{s'_m}. \tag{2}$$

All known cases for $n \leq 5$ (see [12] for detailed proofs in the cases $n \leq 4$) show that the inversion formulas in (2) can be applied to all signatures (p, q) such that $p+q$ is the fixed dimension n. Furthermore, it is easy to check explicitly, for example, that the inversion formula for $n = 5$ also yields the inverses for $n = 4, 3, 2, 1$ as well, i.e., each formula for larger n *automatically gives inverses for all lower dimensions* $m \leq n$ *and for all signatures*. Therefore, we *conjecture* that the explicit formulas for the determinant norm (1) are signature independent, and that, when restricted to lower dimension vector spaces, they yield determinant norms for lower dimensions, generally raised in some power, which can be easily determined from the dimension of a matrix representation (see Example 5 below). The known cases also suggest that formulas (2) always ensure that the inversion commutes with all three main involutions: the reversion, the grade involution and the Clifford conjugation, i.e. $\widetilde{\mathsf{A}^{-1}} = (\widetilde{\mathsf{A}})^{-1}$, $\widehat{\mathsf{A}^{-1}} = (\widehat{\mathsf{A}})^{-1}$ and $\widehat{\widetilde{\mathsf{A}^{-1}}} = (\widehat{\widetilde{\mathsf{A}}})^{-1}$. This is not generally true for an arbitrary involution, for example, for an arbitrary combination of grade negations: $(\mathsf{A}^{-1})_{\bar{i},\ldots} \neq (\mathsf{A}_{\bar{i},\ldots})^{-1}$.

It is well known that Clifford algebras are isomorphic to algebras of square matrices, where left and right inverses coincide. The Clifford MV, therefore, has only one inverse which, depending on our needs, can be written

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[1] In the literature, the GA terminology on MV norms is quite confusing. The book [10] and most of introductory GA lecture courses, for example [4], typically define the MV norm $|\mathsf{A}|^2$ as the scalar part of the product $\big\langle \mathsf{A}\widetilde{\mathsf{A}}\big\rangle_0$, which can be negative. (A better name would be the pseudonorm). If its sign is positive this kind of norm is also called the magnitude. The same term sometimes is used to define a positive square root of absolute value of $\sqrt{\mathrm{abs}(|\mathsf{A}|^2)}$. On the other hand, in [13] a different scalar (named MV norm) is defined which coincides with our determinant norm s_m for $n \leq 5$, whereas in [19] a similar construction is called "the discriminant". In [6,18] this construction has been named "the determinant", because the determinant of a matrix representation of A always coincides with this scalar. Other authors [12] avoid confusion by calling it just "a real scalar". In order to maintain the analogy with MV norm defined in [13] and to distinguish it from the norm definition of [10], we will use the prefixed form "determinant (pseudo)norm" or the shorter form "determinant norm", or just "determinant" if it is clear from the context what is meant. In general, the last term should not be confused with the determinant of a matrix that represents the MV A. It should be noted, however, that the term "determinant" in GA literature is usually employed in the context of linear GA transformations (see, for example, [7], p. 108, the subsection *The determinant*), and that its definition and meaning are related to the exterior product of basis vectors and has nothing to do neither with the above determinant norm nor with a determinant of matrix that represents A.

using either of the two determinant norm forms (1). As a result we have that $\mathsf{A}\mathsf{A}^{-1} = \mathsf{A}^{-1}\mathsf{A} = 1$ which can also serve as a test of correctness of the GA inversion algorithm.

Two-dimensional quadratic space Since the one-dimensional case is trivial we start with the algebras $Cl_{2,0}$, $Cl_{1,1}$ and $Cl_{0,2}$. For $n=2$ let it be $Cl_{2,0}$. Writing $\mathsf{A} = \sum_{J=0}^{2^n-1} a_J \mathbf{e}_J$, where the multi-index J covers all orthonormal basis elements arranged in the increasing order of the degrees followed by the lexicographic order, i.e., the lowest grade elements appear first[2] while elements of the same grade are ordered lexicographically. In the sum the multi-index takes the following explicit values $J = [\{\}, \{1\}, \{2\}, \{1,2\}]$, which illustrate inverse degree lexicographic ordering [5]. We can check that the self-product $\mathsf{A}\mathsf{A}_{\bar{1},\bar{2}}$ immediately gives the required scalar $s_2 = a_{\{\}}^2 - a_1^2 - a_2^2 + a_{1,2}^2$. In the standard notation it is more convenient to rewrite the coefficient indices as in $s_2 = a_0^2 - a_1^2 - a_2^2 + a_3^2$, where the scalar coefficient is indexed by zero. The indices of coefficients for vector components run from 1 to n, while the numerical indices for higher grades rise up to $2^n - 1$. In general, when programming, for a coefficient in front of a basis element of grade-r it is convenient to start the element index enumeration by $\sum_{k=0}^{r-1}\binom{n}{k}$ and to end by $\left(\sum_{k=0}^{r-1}\binom{n}{k}\right) + \binom{n}{r} - 1$. Here $\binom{n}{k}$ denotes the binomial coefficient. For example, for $Cl_{3,0}$, which has three vectors and three bivectors, these expressions give 1 and 3 for first/last vector indices, and 4 and 6 for first/last bivector indices. Taking into account that binomial coefficients with negative indices vanish, this convention enumerates all coefficients of MV and allows an easy transition to more standard notation in a consistent way, for example, for the initial MV we write $\mathsf{A} = a_0 + a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_{12}$ and for the grade-negated MV $\mathsf{A}_{\bar{1},\bar{2}} = a_0 - a_1\mathbf{e}_1 - a_2\mathbf{e}_2 - a_3\mathbf{e}_{12}$. This notation will be used throughout the paper. Note, that indices of basis elements will never be renamed and always be referred to as multi-indices. Before going to higher dimensional algebras a few comments are in place here. The comments are of common character and can be easily generalized to higher dimensional algebras.

1. The explicit form of determinant s_m comprises all coefficients a_i of a multivector A.
2. The condition for a MV to have an inverse is the non-vanishing of its determinant: $s_m \neq 0$. If some intermediate result in the calculation of s_m happens to be zero the entire determinant turns out to zero automatically. This explicit statement was given in order to resolve indeterminate cases like zero divided by zero.
3. Because the reversion leaves the scalar (determinant norm) invariant, the above formula for $n=2$ can be rewritten as $s_2 = \mathsf{A}\mathsf{A}_{\bar{1},\bar{2}} = \mathsf{A}_{\bar{1},\bar{2}}\mathsf{A} = \widetilde{\mathsf{A}}_{\bar{1},\bar{2}}\widetilde{\mathsf{A}} = \widetilde{\mathsf{A}}\widetilde{\mathsf{A}}_{\bar{1},\bar{2}}$ for an arbitrary multivector A. The right hand side of equality, therefore, can be understood as the determinant norm of a multivector $\widetilde{\mathsf{A}}$ written in the right hand side form (where $\widetilde{\mathsf{A}}$ now stands in the

[2]As known, the summation is orderless with respect to basis elements. Nevertheless, it is convenient in advance to settle some order which is required if we want to enumerate coefficients a_i in front of basis elements in a unique way.

right most position). Due to this arbitrariness, inversions can always be written in two forms: $\mathsf{A}^{-1} = \mathsf{A}_{\bar{1},\bar{2}}/s_2 = \mathsf{A}_{\bar{1},\bar{2}}/(\mathsf{A}\mathsf{A}_{\bar{1},\bar{2}}) = \mathsf{A}_{\bar{1},\bar{2}}/(\mathsf{A}_{\bar{1},\bar{2}}\mathsf{A})$ and, correspondingly, $\widetilde{\mathsf{A}^{-1}} = \widetilde{\mathsf{A}}_{\bar{1},\bar{2}}/(\widetilde{\mathsf{A}}_{\bar{1},\bar{2}}\widetilde{\mathsf{A}}) = \widetilde{\mathsf{A}}_{\bar{1},\bar{2}}/(\widetilde{\mathsf{A}}\widetilde{\mathsf{A}}_{\bar{1},\bar{2}}) = \mathsf{A}_{\bar{1},\bar{3}}/(\mathsf{A}_{\bar{2},\bar{3}}\mathsf{A}_{\bar{1},\bar{3}}) = \mathsf{A}_{\bar{1},\bar{3}}/(\mathsf{A}_{\bar{1},\bar{3}}\mathsf{A}_{\bar{2},\bar{3}}) = \widetilde{\mathsf{A}}^{-1}$, where we have explicitly used $\widetilde{\mathsf{A}} = \mathsf{A}_{\bar{2},\bar{3}}$. Determinant norms of A and $\widetilde{\mathsf{A}}$ are equal [18], because the matrix representation of the reversed MV yields the same determinant. Construction of matrix operation that corresponds to the MV reversion is described in [1] for every signature. Since $\mathsf{A}_{\bar{1},\bar{2}} = \widehat{\widetilde{\mathsf{A}}}$, the negated MVs can be expressed through standard involutions when $n = 2$.

4. The determinant expression for $n = 2$ contains only two MVs. From the 8-periodicity table [11] it follows that the algebras $Cl_{2,0}$, $Cl_{1,1}$ and $Cl_{0,2}$ are isomorphic to the algebra $\mathbb{R}(2)$ of real 2×2 matrices or to the algebra $\mathbb{H}$ of quaternions. The determinant of these matrices is a quadratic polynomial in the coefficients a_i of the MVs; therefore, this polynomial can be constructed by multiplying just two MVs. Below we shall see that in all cases the total polynomial degree of the matrix determinant (where the coefficients a_i play the role of variables) always matches the number of MVs in the determinant product. The determinant of a matrix representation with quaternionic entries can be calculated using an isomorphism $\mathbb{H} \cong \mathbb{C}(2)$, i.e. we first replace quaternions by 2×2 block matrices and then calculate the determinant (the real scalar) of the resulting matrix. This practical procedure allows us to avoid considering numerous definitions of the determinant of $\mathbb{H}(n)$ due to non-commutativity.

Three-dimensional quadratic space Our goal is to eliminate as many grades as possible by forming suitable self-products until finally the grade-0 element (determinant norm) is left. As another example, let us calculate inverses in the algebra $Cl_{2,1}$. It is easy to check that the product of $\mathsf{A} = a_0 + a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + a_3\mathbf{e}_3 + a_4\mathbf{e}_{12} + a_5\mathbf{e}_{13} + a_6\mathbf{e}_{23} + a_7\mathbf{e}_{123}$ with $\mathsf{A}_{\bar{1},\bar{2}}$ contains no term of grades 1 and 2. The result is a new multivector $\mathsf{B} = \mathsf{A}\mathsf{A}_{\bar{1},\bar{2}} = b_0 + b_7\mathbf{e}_{123}$ which consists of a scalar and a grade-3 element with coefficients $b_0 = a_0^2 - a_1^2 - a_2^2 + a_3^2 + a_4^2 - a_5^2 - a_6^2 + a_7^2$ and $b_7 = -2a_3a_4 + 2a_2a_5 - 2a_1a_6 + 2a_0a_7$. Repeating grade negation procedure one finds that the grade-3 part can be removed too, and we obtain the determinant norm $s_4 = \mathsf{B}\mathsf{B}_{\bar{3}} = \mathsf{A}\mathsf{A}_{\bar{1},\bar{2}}(\mathsf{A}\mathsf{A}_{\bar{1},\bar{2}})_{\bar{3}} = b_0^2 - b_7^2$. Reversion of s_4 then immediately yields an alternative form of the norm $s_4' = (\widetilde{\mathsf{A}}_{\bar{1},\bar{2}}\widetilde{\mathsf{A}})_{\bar{3}}\widetilde{\mathsf{A}}_{\bar{1},\bar{2}}\widetilde{\mathsf{A}}$. Of course, with different signatures, the same formula will give distinct real expressions, which differ in signs at individual coefficients a_i. The important thing is that despite the fact that the scalar s_4' was calculated in the algebra $Cl_{2,1}$, exactly the same sequence of products and grade negations will produce the scalar (generally different) in all other algebras with $n = 3$. Also note, that the determinant norm in this case is determined by two terms, b_0^2 and b_7^2, the difference of which should not be equal to zero for the MV A to be invertible. The condition $b_0^2 - b_7^2 \neq 0$ exactly matches the MV invertibility condition, which according to the 8-periodicity table may be obtained by calculating a matrix representation of the MV.

Four-dimensional quadratic space In $n=4$ case, in trying to eliminate as many grades as possible we can proceed in two alternative ways. Firstly, we can negate simultaneously the grades 1 and 2 and then in the next step the grades 3 and 4. Alternatively, we can eliminate the grades 2 and 3, and then 1 and 4. Both choices are valid. However, if we choose 1 and 3, and then 2 and 4 grade combinations, neither one will do the job. Thus, we find the following formulas for determinant norm $s_4 = \mathsf{A}\mathsf{A}_{\bar{1},\bar{2}}(\mathsf{A}\mathsf{A}_{\bar{1},\bar{2}})_{\bar{3},\bar{4}} = \mathsf{A}\mathsf{A}_{\bar{2},\bar{3}}(\mathsf{A}\mathsf{A}_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}$ and $s_4' = (\widetilde{\mathsf{A}}_{\bar{1},\bar{2}}\widetilde{\mathsf{A}})_{\bar{3},\bar{4}}\,\widetilde{\mathsf{A}}_{\bar{1},\bar{2}}\widetilde{\mathsf{A}} = (\widetilde{\mathsf{A}}_{\bar{2},\bar{3}}\widetilde{\mathsf{A}})_{\bar{1},\bar{4}}\widetilde{\mathsf{A}}_{\bar{2},\bar{3}}\widetilde{\mathsf{A}}$. It is easy to check that both expressions indeed give determinant norms. The occurrence of the symbol A as often as four times in the products s_4 and s_4' is again what we expect from the matrix representations for $n=4$. The total degree of the determinant, considered as a polynomial function of the coefficients of the MV, is 4 for all algebras in the case $n=4$. Despite different forms, the expanded explicit expressions for s_m and s_m' were found to be equal, $s_m = s_m'$, as it should be [12]. Our computations show that the above formulas for s_m and s_m' are the only possible ways to get the determinant norm in the case $n=4$ by using geometric product and negation operations.

Five-dimensional quadratic space This is the largest dimension, $n=5$, when consecutive grade elimination works by factorizing the determinant norm into the product of the initial MV and negated ones. The grade elimination sequence is similar [12]: (1) eliminate simultaneously grades 2 and 3; (2) then eliminate grades 1 and 4; (3) finally eliminate grade 5. Apart from the last step this sequence is exactly the same as the second alternative of the case $n=4$. The final result is $s_8' = (\mathsf{F}\widetilde{\mathsf{A}})_{\bar{5}}\mathsf{F}\widetilde{\mathsf{A}}$ with $\mathsf{F} = (\widetilde{\mathsf{A}}_{\bar{2},\bar{3}}\widetilde{\mathsf{A}})_{\bar{1},\bar{4}}\widetilde{\mathsf{A}}_{\bar{2},\bar{3}}$ and $s_8 = \mathsf{A}\mathsf{D}\,(\mathsf{A}\mathsf{D})_{\bar{5}}$ with $\mathsf{D} = \mathsf{A}_{\bar{2},\bar{3}}(\mathsf{A}\mathsf{A}_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}$.

The total degree of determinant polynomial in this case is 8, which again exactly matches the number of MVs in the determinant norm product. With our program [2] we have found that the determinant norm s_8 can be written in 52 different ways as presented in Table 1.

Example 1. Let's take $\mathsf{A} = 3 + \mathbf{e}_2 + \mathbf{e}_5 - \mathbf{e}_{12} - \mathbf{e}_{15} + 3\mathbf{e}_{125}$ in $Cl_{4,1}$. It is easy to check that $\mathsf{A}\mathsf{A}_{\bar{2},\bar{3}} = 0$. In Table 1, all formulas that contain $H = \mathsf{A}\mathsf{A}_{\bar{2},\bar{3}}$ immediately allow to conclude that the determinant norm of this MV is zero (see entries $N = 13, 14, 15, 17, 18$) and, therefore, the inverse of A does not exist. Now let's try to find the determinant of A with a formula that does not contain $\mathsf{A}\mathsf{A}_{\bar{2},\bar{3}} = 0$, for example with $H' = \mathsf{A}\mathsf{A}_{\bar{1},\bar{2},\bar{5}}$ (see the first line and $N = 15, 16$ in Table 1), from which the final result is not so obvious. First, we calculate $H' = 18 + 18\mathbf{e}_{125}$, which is not zero. However, computing the next step $H'(H')_{\bar{3}}$ and $(H')_{\bar{3}}H'$ in the last two lines in Table 1 we get zero again.

Example 2. Given $\mathsf{A} = 1 + 2\mathbf{e}_1 + 3\mathbf{e}_{23} + 4\mathbf{e}_{2345}$ in $Cl_{5,0}$ let us find A^{-1} using a couple of alternative formulas. First, we shall use the new computationally efficient formula $D_1 = HH_{\bar{1},\bar{5}}(HH_{\bar{1},\bar{5}})_{\bar{3},\bar{4}}$ (underlined in Table 1). The computation of H yields $H = \mathsf{A}\mathsf{A}_{\bar{2},\bar{3}} = 30 + 4\mathbf{e}_1 + 8\mathbf{e}_{2345} + 16\mathbf{e}_{12345}$. Then $HH_{\bar{1},\bar{5}} = 692 + 352\mathbf{e}_{2345}$. And lastly, $D_1 = 354960$. Then the inverse

Table 1. Alternative formulas for MV determinant norm for GAs of spaces of dimension $n = 5$ listed according to the number of negations

$N\downarrow$	Abbreviation $\rightarrow$	$H = \mathsf{A}\widetilde{\mathsf{A}} = \mathsf{A}\mathsf{A}_{\bar2,\bar3}$	$H' = \mathsf{A}\widehat{\widetilde{\mathsf{A}}} = \mathsf{A}\mathsf{A}_{\bar1,\bar2,\bar5}$
13	$H(H(H_{\bar1,\bar4}H)_{\bar5})_{\bar1,\bar4}$	$H(H(HH_{\bar1,\bar4})_{\bar5})_{\bar1,\bar4}$	$H(H(HH_{\bar1,\bar5})_{\bar4})_{\bar1,\bar5}$
	$H(H(HH_{\bar4,\bar5})_{\bar1})_{\bar4,\bar5}$	$HH_{\bar1,\bar4}(H_{\bar1,\bar4}H)_{\bar5}$	$\underline{HH_{\bar1,\bar4}(HH_{\bar1,\bar4})_{\bar5}}$
	$HH_{\bar1,\bar5}(H_{\bar1,\bar5}H)_{\bar4}$	$HH_{\bar4,\bar5}(H_{\bar4,\bar5}H)_{\bar1}$	
14	$H(H_{\bar1,\bar4}HH_{\bar1,\bar5})_{\bar4,\bar5}$	$H(H_{\bar1,\bar4}HH_{\bar4,\bar5})_{\bar1,\bar5}$	$H(H_{\bar1,\bar5}H_{\bar1,\bar4}H)_{\bar4,\bar5}$
	$H(H_{\bar1,\bar5}H_{\bar4,\bar5}H))_{\bar1,\bar4}$	$H(H_{\bar1,\bar5}HH_{\bar1,\bar4})_{\bar4,\bar5}$	$H(H_{\bar4,\bar5}H_{\bar1,\bar4}H)_{\bar1,\bar5}$
	$H(H_{\bar4,\bar5}H_{\bar1,\bar5}H)_{\bar1,\bar4}$	$H(H_{\bar4,\bar5}HH_{\bar1,\bar4})_{\bar1,\bar5}$	$H(H(H_{\bar1,\bar5}H)_{\bar3,\bar4})_{\bar1,\bar5}$
	$H(H(H_{\bar4,\bar5}H)_{\bar1,\bar3})_{\bar4,\bar5}$	$H(HH_{\bar1,\bar4}H_{\bar1,\bar5})_{\bar4,\bar5}$	$H(HH_{\bar1,\bar4}H_{\bar4,\bar5})_{\bar1,\bar5}$
	$H(HH_{\bar1,\bar5}H_{\bar4,\bar5})_{\bar1,\bar4}$	$H(HH_{\bar4,\bar5}H_{\bar1,\bar5})_{\bar1,\bar4}$	$HH_{\bar1,\bar4}H_{\bar1,\bar5}H_{\bar4,\bar5}$
	$HH_{\bar1,\bar4}H_{\bar4,\bar5}H_{\bar1,\bar5}$	$\underline{HH_{\bar1,\bar5}(HH_{\bar1,\bar5})_{\bar3,\bar4}}$	$HH_{\bar1,\bar5}H_{\bar4,\bar5}H_{\bar1,\bar4}$
	$\underline{HH_{\bar4,\bar5}(HH_{\bar4,\bar5})_{\bar1,\bar3}}$	$HH_{\bar4,\bar5}H_{\bar1,\bar5}H_{\bar1,\bar4}$	
15	$H(H_{\bar1,\bar4}(H_{\bar1,\bar5}H)_{\bar3})_{\bar4,\bar5}$	$H(H_{\bar1,\bar4}(H_{\bar4,\bar5}H)_{\bar3})_{\bar1,\bar5}$	$H(H_{\bar1,\bar5}(HH_{\bar4,\bar5})_{\bar3})_{\bar1,\bar4}$
	$H(H_{\bar4,\bar5}(HH_{\bar1,\bar5})_{\bar3})_{\bar1,\bar4}$	$H(H(H_{\bar1,\bar5}H_{\bar1,\bar4})_{\bar3})_{\bar4,\bar5}$	$H(H(H_{\bar4,\bar5}H_{\bar1,\bar4})_{\bar3})_{\bar1,\bar5}$
	$HH_{\bar1,\bar5}(H_{\bar1,\bar4}H_{\bar4,\bar5})_{\bar3}$	$HH_{\bar4,\bar5}(H_{\bar1,\bar4}H_{\bar1,\bar5})_{\bar3}$	
17	$H(H_{\bar1,\bar4}(H_{\bar1,\bar4}H_{\bar1,\bar5})_{\bar1})_{\bar1,\bar5}$	$H(H_{\bar1,\bar4}(H_{\bar1,\bar4}H_{\bar4,\bar5})_{\bar4})_{\bar4,\bar5}$	$H(H_{\bar1,\bar5}(H_{\bar1,\bar5}H_{\bar1,\bar4})_{\bar1})_{\bar1,\bar4}$
	$H(H_{\bar1,\bar5}(H_{\bar1,\bar5}H_{\bar4,\bar5})_{\bar5})_{\bar4,\bar5}$	$H(H_{\bar1,\bar5}(H_{\bar4,\bar5}H_{\bar1,\bar5})_{\bar5})_{\bar4,\bar5}$	$H(H_{\bar4,\bar5}(H_{\bar1,\bar5}H_{\bar4,\bar5})_{\bar5})_{\bar1,\bar5}$
	$H(H_{\bar4,\bar5}(H_{\bar4,\bar5}H_{\bar1,\bar4})_{\bar4})_{\bar1,\bar4}$	$H(H_{\bar4,\bar5}(H_{\bar4,\bar5}H_{\bar1,\bar5})_{\bar5})_{\bar1,\bar5}$	
18	$H(H_{\bar1,\bar4}(H_{\bar1,\bar5}H_{\bar1,\bar4})_{\bar1,\bar3})_{\bar1,\bar5}$	$H(H_{\bar1,\bar4}(H_{\bar4,\bar5}H_{\bar1,\bar4})_{\bar3,\bar4})_{\bar4,\bar5}$	$H(H_{\bar1,\bar5}(H_{\bar1,\bar4}H_{\bar1,\bar5})_{\bar1,\bar3})_{\bar1,\bar4}$
	$H(H_{\bar4,\bar5}(H_{\bar1,\bar4}H_{\bar4,\bar5})_{\bar3,\bar4})_{\bar1,\bar4}$		
15	$H'(H'(H'H'_{\bar3})_{\bar4})_{\bar3}$	$H'H'_{\bar3}(H'_{\bar3}H')_{\bar4}$	
16	$H'(H'(H'_{\bar3}H')_{\bar1,\bar4})_{\bar3}$	$H'(H'_{\bar3}(H'H'_{\bar3})_{\bar1,\bar4}$	

For example, the number of negations N in the first line is determined by four $H = \mathsf{A}\mathsf{A}_{\bar2,\bar3}$, each including 2 negations, $(\bar2, \bar3)$, and five explicit negations $(\bar1, \bar4) + \bar5 + (\bar1, \bar4)$ in the formula, resulting in $2 * 4 + 5 = 13$ total negations. Computationally preferred forms that contain the largest number of repeated factors are underlined

is $\mathsf{A}^{-1} = \frac{\mathsf{A}_{\bar2,\bar3}H_{\bar1,\bar5}(HH_{\bar1,\bar5})_{\bar3,\bar4}}{D_1} = \frac{1}{354960}(3576 + 96\mathbf{e}_1 - 53832\mathbf{e}_{23} - 15072\mathbf{e}_{45} - 8592\mathbf{e}_{123} - 28992\mathbf{e}_{145} + 47424\mathbf{e}_{2345} - 8256\mathbf{e}_{12345})$.

Now let's compute the determinant norm of A using $D_2 = HH_{\bar4,\bar5}H_{\bar1,\bar5}H_{\bar1,\bar4}$, which computationally is less efficient, because it contains a smaller number of repeated factors. The computation of $HH_{\bar4,\bar5}$ yields $596 - 16\mathbf{e}_1$. Then, $HH_{\bar4,\bar5}H_{\bar1,\bar5}$ gives $17944 - 2864\mathbf{e}_1 + 5024\mathbf{e}_{2345} - 9664\mathbf{e}_{12345}$. Finally, $D_2 = 354960$. Of course, these formulas give the same explicit expression of the inverse MV in symbolic form as well.

Can the above described determinant computation procedure be extended beyond $n = 5$? The short answer is 'yes' if, as shown below, we allow linear combinations of grade-negated self-products with properly chosen numerical coefficients. Before describing this case let us derive some useful formulas for the even subalgebra of $n \le 6$. The even subalgebras are directly related with the spinor groups that are very important in quantum mechanics [20].

Table 2. Explicit inversion formulas for even MVs in a coordinate-free form for Clifford algebras of spaces of dimension $n=p+q\le 6$

$Cl_{p,q}$	$\left(\mathsf{A}^+\right)^{-1}$	
$p+q=2$	$\frac{B\left(\mathsf{A}^+B\right)}{\left(\mathsf{A}^+B\right)^2}$,	$B=(v)_{\bar{1}}=-v$
$p+q=3$	$\frac{C\left(\mathsf{A}^+C\right)}{\left(\mathsf{A}^+C\right)^2}$,	$C=(\mathsf{A}^+v)_{\bar{1}}$
$p+q=4$	$\frac{C\left(\mathsf{A}^+C\right)}{\left(\mathsf{A}^+C\right)^2}$,	$C=(\mathsf{A}^+v)_{\bar{1}}$
$p+q=5$	$\frac{D\left(\mathsf{A}^+D\right)}{\left(\mathsf{A}^+D\right)^2}$,	$D=(\mathsf{A}^+v)_{\bar{3}}\left(\mathsf{A}^+(\mathsf{A}^+v)_{\bar{3}}\right)_{\bar{1}}$
$p+q=6$	$\frac{D\left(\mathsf{A}^+D\right)_{\bar{6}}}{\left(\mathsf{A}^+D\right)\left(\mathsf{A}^+D\right)_{\bar{6}}}$,	$D=(\mathsf{A}^+v)_{\bar{3}}\left(\mathsf{A}^+(\mathsf{A}^+v)_{\bar{3}}\right)_{\bar{1}}$

The quantities $\left(\mathsf{A}^+B\right)^2/v^2$, $\left(\mathsf{A}^+C\right)^2/v^2$, $\left(\mathsf{A}^+D\right)^2/v^4$ and $\left(\mathsf{A}^+D\right)\left(\mathsf{A}^+D\right)_{\bar{6}}/v^4$ are the determinant norms of the respective MVs. Here, the nonisotropic and not necessarily normalized vector v can be replaced by any vector $\mathbf{e}_i$ of the orthonormal basis for computational efficiency

3. Inverses of Even Multivectors

When a MV consists of even grade elements only, i.e. $\mathsf{A}^+ = \langle\mathsf{A}\rangle_0 + \langle\mathsf{A}\rangle_2 + \langle\mathsf{A}\rangle_4 + \cdots$, simpler formulas for inverse MVs can be constructed as shown in Table 2. The nonisotropic and not necessarily normalized vector v in these formulas plays the role of a dummy variable.

It is interesting to observe that in contrast to the general case considered in the next section, the inverse of even MV (see Table 2) may be a single self-negated product (as in Table 3 for $n=5$) and not a linear combination of such products. This property is to be expected, because there exists well-known isomorphisms between the even Clifford of a space of dimension $n+1$ and the full Clifford algebra of a space of dimension n.

$$Cl^+_{p,q+1}\cong Cl_{p,q},\quad Cl^+_{p+1,q}\cong Cl_{q,p}\,. \tag{3}$$

Because we can write the inverse MV as a self-negated product when $n=5$ (see Table 3), it is not surprising that according to isomorphisms (3) we can do the same for an even MV when $n=6$. Unfortunately, as we shall see in Sect. 4 this property does not extend to full Clifford algebras when $n=6$.

4. Inverse of a General MV When $n=6$

4.1. Insufficiency of Single Self-Negated Products in the Case $n=6$

Let us take the algebra $Cl_{6,0}$. Using *Mathematica* GA package [2], after some experimentation with a pair of self-negated products derived from a general MV, it is not difficult to ascertain that we can eliminate simultaneously either the grades 1, 2, 5 and 6 (then the grades 0, 3 and 4 survive) or, alternatively, the grades 2, 3 and 6 (then the grades 0, 1, 4 and 5 survive). In both cases grade 4 remains and, therefore, cannot be eliminated by the method of simple self-negated product used till now. This conclusion strictly follows from an attempt to simultaneously nullify the coefficients corresponding to all grade-4 elements of the basis by using all $2^6 = 64$ possible combinations of grade negations in the two factors of the product. The grade 4 therefore is distinct from the other grades and deserves special examination. From experiments by computer it turns out that, in the subalgebra with basis $\{1, \mathbf{e}_{1256}, \mathbf{e}_{1346}, \mathbf{e}_{2345}\}$, the self-product of any multivector $\mathsf{B} = a_0 + a_{47}\mathbf{e}_{1256} + a_{49}\mathbf{e}_{1346} + a_{52}\mathbf{e}_{2345}$ by any grade negation of B yields a new MV that has at least one non-zero component of grade 4 in this base. This is the reason why such a single self-product fails in eliminating the grade-4 part and, as we shall see, one has to use a combination of at least a pair of self-products.

4.2. Linear Combination of Self-Products

Before considering the general case, we shall note that the determinant of a matrix representation (computed using symbolic coefficients) of a restricted MV, that consists of just a scalar and a general grade-4 element, can be written as the square of some polynomial s_4 in the coefficients of the MV, i.e. as $\det\big(\langle\mathsf{A}\rangle_0 + \langle\mathsf{A}\rangle_4\big) = s_8 = (s_4)^2$. Thus, we assume that one can always extract the square root of s_8. From this follows that in search of the inverse of $\langle\mathsf{A}\rangle_0 + \langle\mathsf{A}\rangle_4$, as a first step one can try to test only linear combinations of products of four negated MVs instead of eight as required in the general case. Due to the above arguments, we assume that the square root of the determinant norm may be written as a linear combination of the following form

$$\begin{aligned} &s_{4f} + s_{4g} \\ &\quad = b_1 B f_5\Big(f_4(B) f_3\big(f_2(B) f_1(B)\big)\Big) + b_2 B g_5\Big(g_4(B) g_3\big(g_2(B) g_1(B)\big)\Big), \end{aligned} \tag{4}$$

where each of f_j and g_j is either the identity mapping or the grade-4 negation, and b_1, b_2 are unknown scalar coefficients of the linear combination.

Once a pattern of linear combination for the square root of the determinant norm has been fixed, we can calculate an explicit symbolic form of matrix representation of the above mentioned MV $\mathsf{B} = a_0 + a_{47}\mathbf{e}_{1256} + a_{49}\mathbf{e}_{1346} + a_{52}\mathbf{e}_{2345}$, then compute the matrix determinant, take its square root and compare it with the GA expression (4) after the same GA multivector B has been inserted.

The negation functions f_j and g_j that control the signs of the grades can be modelled as multiplication by unknown coefficients p_{4jk}, where the index j denotes the number of the involution f_j or g_j in the self-product and k is the number of the term in the linear combination (when $n = 6$, $k = 1$ for f and $k = 2$ for g). Later, when we shall deal with negations of other grades the first index i in p_{ijk} will indicate a possibly negated grade-i. For the moment the index i is fixed to 4, which corresponds to the current grade of B (we will ignore negations of scalar, because it is equivalent to negation of all other remaining MV grades). The coefficients p_{ijk} take only the values

± 1, where -1 means that the involution changes the sign of the grade i, while $+1$ means that it is the identity map.

In the case $n = 6$ under consideration, comparison with the square root of the determinant of a matrix representation of B yields a system of four equations for each basis element (including the scalar) of B. The system is too long to be fully presented here, therefore a small characteristic part of it is written down in a truncated form below,

$$\begin{cases} a_0^4 - 2a_0^2a_{47}^2 + b_2a_0^2a_{47}^2p_{412}p_{422} + < 72 \text{ monomials} >= 0 \\ b_1a_0^3a_{52} + 2b_2a_0a_{47}^2a_{52}p_{412}p_{422}p_{432}p_{442}p_{452} + < 72 \text{ monomials} >= 0 \\ < 74 \text{ monomials} >= 0 \\ < 74 \text{ monomials} >= 0 \end{cases} \tag{5}$$

We see that even for a simple B which contains only 4 components (with the basis elements 1, $\mathbf{e}_{1256}$, $\mathbf{e}_{1346}$ and $\mathbf{e}_{2345}$) the system is highly nonlinear in p_{4ij}. We can, however, try to substitute concrete values for p_{4ij} one by one to get much simpler systems that contain only variables b_1, b_2 and a_i. Then we can try to solve each simple system separately with respect to b_1, b_2 for arbitrary coefficients a_i. Only few of them have solutions. In fact, we have solved the systems with *Mathematica* command **SolveAlways**[ ]. After testing all $2^{10} = 1024$ possible values of p_{4ij} we have found two sets of solutions, $\{b_1 = -2/3,\ b_2 = -1/3,\ p_{411} = -1,\ p_{412} = 1,\ p_{421} = -1,\ p_{422} = 1,\ p_{431} = -1,\ p_{432} = -1,\ p_{441} = -1,\ p_{442} = 1,\ p_{451} = -1,\ p_{452} = 1\}$ and $\{b_1 = -1/3,\ b_2 = -2/3,\ p_{411} = 1,\ p_{412} = -1,\ p_{421} = 1,\ p_{422} = -1,\ p_{431} = -1,\ p_{432} = -1,\ p_{441} = 1,\ p_{442} = -1,\ p_{451} = 1,\ p_{452} = -1\}$. The obtained solution is unique up to the permutation of the two terms, and also up to a factor ± 1; indeed, if we replace (b_1, b_2) with $(-b_1, -b_2)$, we also obtain a square root of the determinant norm.

It is easy to check that the above solution (though computed for a highly simplified MV, namely, a scalar plus three grade-4 base elements) works flawlessly for a more general MV (a scalar plus any number of grade-4 elements) $\mathsf{C} = a_0 + a_{42}\mathbf{e}_{1234} + a_{43}\mathbf{e}_{1235} + a_{44}\mathbf{e}_{1236} + a_{45}\mathbf{e}_{1245} + a_{46}\mathbf{e}_{1246} + a_{47}\mathbf{e}_{1256} + a_{48}\mathbf{e}_{1345} + a_{49}\mathbf{e}_{1346} + a_{50}\mathbf{e}_{1356} + a_{51}\mathbf{e}_{1456} + a_{52}\mathbf{e}_{2345} + a_{53}\mathbf{e}_{2346} + a_{54}\mathbf{e}_{2356} + a_{55}\mathbf{e}_{2456} + a_{56}\mathbf{e}_{3456}$. This is what one expects, because grade involution acts on all components having the same grade similarly. Starting from a simple MV and then augmenting it until the number of solutions could not be reduced further allows one to keep the whole calculation size manageable. This considerably speeds up the search procedure for involutions and balances calculation complexity against speed.

Once f_j, g_j and the coefficients b_k in Eqs. (4) and (5) have been determined, we can renew the search for other negation involutions in exactly the same way. Our next task is to double the number of multivectors in the products (4), because we know that the determinant of a general MV requires 8 multipliers in order to match the total degree polynomial of the determinant of a matrix representation.

It was already mentioned that in the product $\mathsf{AA}_{i,j,\ldots}$ one can simultaneously eliminate either the grades 1, 2, 5 and 6 (then the grades 0, 3 and 4 remain), or, alternatively, the grades 2, 3 and 6 (then the grades 0, 1, 4

TABLE 3. Summary of formulas for inverse multivectors A^{-1} in Clifford algebras of dimension $n \leq 6$

$Cl_{p,q}$	A^{-1}	
$p+q=0$	$\frac{\mathsf{B}}{\mathsf{AB}}$,	$\mathsf{B}=1$
$p+q=1$	$\frac{\mathsf{B}(\mathsf{AB})_{\bar{1}}}{\mathsf{AB}\,(\mathsf{AB})_{\bar{1}}}=\frac{\widehat{\mathsf{A}}}{\mathsf{A}\widehat{\mathsf{A}}}$,	$\mathsf{B}=1$
$p+q=2$	$\frac{\mathsf{C}}{\mathsf{AC}}=\frac{\widehat{\widetilde{\mathsf{A}}}}{\mathsf{A}\widehat{\widetilde{\mathsf{A}}}}$,	$\mathsf{C}=\mathsf{A}_{\bar{1},\bar{2}}$
$p+q=3$	$\frac{\mathsf{C}(\mathsf{AC})_{\bar{3}}}{\mathsf{AC}\,(\mathsf{AC})_{\bar{3}}}=\frac{\widehat{\widetilde{\mathsf{A}}}\widehat{\mathsf{A}}\widetilde{\mathsf{A}}}{\mathsf{A}\widehat{\widetilde{\mathsf{A}}}\widehat{\mathsf{A}}\widetilde{\mathsf{A}}}$,	$\mathsf{C}=\mathsf{A}_{\bar{1},\bar{2}}$
$p+q=4$	$\frac{\mathsf{D}}{\mathsf{AD}}$,	$\mathsf{D}=\mathsf{A}_{\bar{2},\bar{3}}(\mathsf{AA}_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}$
		or $\mathsf{D}=\mathsf{A}_{\bar{1},\bar{2}}(\mathsf{AA}_{\bar{1},\bar{2}})_{\bar{3},\bar{4}}$
$p+q=5$	$\frac{\mathsf{D}(\mathsf{AD})_{\bar{5}}}{\mathsf{AD}\,(\mathsf{AD})_{\bar{5}}}$,	$\mathsf{D}=\mathsf{A}_{\bar{2},\bar{3}}(\mathsf{AA}_{\bar{2},\bar{3}})_{\bar{1},\bar{4}}$
$p+q=6$	$\frac{\mathsf{G}}{\mathsf{AG}}$,	$\mathsf{G}=\frac{1}{3}\mathsf{A}_{\bar{2},\bar{3},\bar{6}}\Big(\mathsf{H}(\mathsf{HH})_{\bar{1},\bar{4},\bar{5}}+2\big(\mathsf{H}_{\bar{4}}(\mathsf{H}_{\bar{4}}\mathsf{H}_{\bar{4}})_{\bar{1},\bar{4},\bar{5}}\big)_{\bar{4}}\Big)$
		or $\mathsf{G}=\frac{1}{3}\mathsf{A}_{\bar{2},\bar{3},\bar{6}}\Big(\big(\mathsf{H}(\mathsf{HH}_{\bar{1},\bar{5}})_{\bar{4}}\big)_{\bar{1},\bar{5}}+2\big(\mathsf{H}_{\bar{4},\bar{5}}(\mathsf{H}_{\bar{4},\bar{5}}\mathsf{H}_{\bar{1},\bar{4}})_{\bar{4}}\big)_{\bar{1},\bar{4}}\Big)$
		with $\mathsf{H}=\mathsf{AA}_{\bar{2},\bar{3},\bar{6}}=\mathsf{A}\widetilde{\mathsf{A}}$

and 5 survive). All in all there are $\frac{6!}{2!4!}+\frac{6!}{3!3!}+1=36$ components of grades 2, 3 and 6 which can be eliminated simultaneously. This is more than 28 components of grades 1, 2, 5 and 6. Therefore, if we replace B in Eq. (4) by the self-negated product $\mathsf{AA}_{\bar{2},\bar{3},\bar{6}}$ we are left to deal with a self-negated product of four multivectors with only components of grades 0, 1, 4, 5, from which we can ignore the grades 0 and 4 for which f_j and g_j have already been established. Thus, repeating the same procedure we can find a number of valid solutions for the coefficients p_{1jk}, p_{2jk} and p_{5jk}. In order to speed up the derivation we have used the multivector $a_0+a_1\mathbf{e}_1+a_{22}\mathbf{e}_{123}+a_{47}\mathbf{e}_{1256}$ first. The obtained result then was explicitly verified (i.e. proved by explicit expansion in orthogonal basis) using the most general $Cl_{6,0}$ multivector with symbolic coefficients. Finally we found 320 valid forms of determinant norm. After removing all superfluous involutions the number of possible determinant norms was reduced to 16+4=20. All these norms are presented in Table 4.

After the determinant norm has been found the inverse multivector can be immediately written using Eq. (2). The results for inverse MVs in the case $n \leq 6$ are summarized in Table 3.

The denominators in Table 3, which are scalars, are equal to the determinant norm of the MV A. At the same time they give the condition of existence of the inverse MV. It has been found that the determinant norm exactly matches the expression of the determinant calculated with matrix representation of the MV (in fact, the whole derivation of the determinant norm formulas relied on this match). All formulas were proved by the explicit

Table 4. The formulas giving the determinant norm of the MV A when $n=6$

$N\downarrow$	Abbreviation $\rightarrow H = A\widetilde{A} = AA_{\bar{2},\bar{3},\bar{6}}$
38	$\frac{1}{3}H(H(HH_{\bar{1},\bar{5}})_{\bar{4}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{1},\bar{4}}(H_{\bar{1},\bar{4}}H_{\bar{4},\bar{5}})_{\bar{4}})_{\bar{4},\bar{5}}$
	$\frac{1}{3}HH_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H)_{\bar{4}} + \frac{2}{3}H(H_{\bar{1},\bar{4}}(H_{\bar{1},\bar{4}}H_{\bar{4},\bar{5}})_{\bar{4}})_{\bar{4},\bar{5}}$
	$\frac{1}{3}H(H(HH_{\bar{1},\bar{5}})_{\bar{4}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{4},\bar{5}}(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{4}})_{\bar{1},\bar{4}}$
	$\frac{1}{3}HH_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H)_{\bar{4}} + \frac{2}{3}H(H_{\bar{4},\bar{5}}(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{4}})_{\bar{1},\bar{4}}$
34	$\frac{1}{3}HH(HH)_{\bar{1},\bar{4},\bar{5}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}}$
36	$\frac{1}{3}HH(HH)_{\bar{1},\bar{4},\bar{5}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{4}})_{\bar{4}}$
	$\frac{1}{3}HH(HH)_{\bar{1},\bar{4},\bar{5}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{4}}H_{\bar{4}})_{\bar{4}})_{\bar{1},\bar{4},\bar{5}}$
	$\frac{1}{3}H(H_{\bar{1},\bar{5}}(HH)_{\bar{4}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}}$
	$\frac{1}{3}HH(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{4}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}}$
38	$\frac{1}{3}H(H_{\bar{1},\bar{5}}(HH)_{\bar{4}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{4}})_{\bar{4}}$
	$\frac{1}{3}HH(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{4}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{4}})_{\bar{4}}$
	$\frac{1}{3}H(H_{\bar{1},\bar{5}}(HH)_{\bar{4}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{4}}H_{\bar{4}})_{\bar{4}})_{\bar{1},\bar{4},\bar{5}}$
	$\frac{1}{3}HH(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{4}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{4}}H_{\bar{4}})_{\bar{4}})_{\bar{1},\bar{4},\bar{5}}$
42	$\frac{1}{3}HH(HH)_{\bar{1},\bar{4},\bar{5}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}}$
	$\frac{1}{3}H(H_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}}$
44	$\frac{1}{3}H(H_{\bar{1},\bar{5}}(HH)_{\bar{4}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}}$
	$\frac{1}{3}HH(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{4}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}}$
	$\frac{1}{3}H(H_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{4}})_{\bar{4}}$
	$\frac{1}{3}H(H_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{4}}H_{\bar{4}})_{\bar{4}})_{\bar{1},\bar{4},\bar{5}}$
50	$\frac{1}{3}H(H_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}}$

The integer N counts the total number of negations in the norm (see Table 1 for explanation)

calculation of the symbolic inverse of the most general MV with several Clifford algebras having different signatures (p,q). For this task we wrote the *Mathematica* package for symbolic calculations in Clifford algebras [2], which also can be found in *Mathematica* package data base http://packagedata.net/. The specific property of our GA program is that it can simultaneously work with a number of Clifford algebras having different signatures (p,q), which was a great help in computer-assisted verifications.

5. Classification and Examples

In Table 4 we have collected all formulas obtained by the described method that represent the determinant norm at $n = 6$ in the pattern (4). All expressions for determinant norm naturally split into two types. The first four formulas at the beginning of Table 4 ($N = 38$) constitute the first type or class. All four expressions from this class can be obtained by forming pairs from two sets

$$\left\{\frac{1}{3}H(H(HH_{\bar{1},\bar{5}})_{\bar{4}})_{\bar{1},\bar{5}},\ \frac{1}{3}HH_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H)_{\bar{4}}\right\}, \quad \text{and}$$
$$\left\{\frac{2}{3}H(H_{\bar{1},\bar{4}}(H_{\bar{1},\bar{4}}H_{\bar{4},\bar{5}})_{\bar{4}})_{\bar{4},\bar{5}}, \frac{2}{3}H(H_{\bar{4},\bar{5}}(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{4}})_{\bar{1},\bar{4}}\right\}$$

in any combination. This gives the first four, $2\times 2 = 4$, formulas. The remaining 16 formulas, which constitute the second class can be obtained similarly by forming all possible pairs from the sets

$$\left\{\frac{1}{3}HH(HH)_{\bar{1},\bar{4},\bar{5}},\ \frac{1}{3}H(H_{\bar{1},\bar{5}}(HH)_{\bar{4}})_{\bar{1},\bar{5}}, \frac{1}{3}HH(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{4}},\right.$$
$$\left.\frac{1}{3}H(H_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{5}}\right\},$$

and

$$\left\{\frac{2}{3}H(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}}, \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{4}}H_{\bar{4}})_{\bar{4}})_{\bar{1},\bar{4},\bar{5}},\right.$$
$$\left.\frac{2}{3}H(H_{\bar{4}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{4}})_{\bar{4}}, \frac{2}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}}\right\}.$$

This yields remaining $4 \times 4 = 16$ formulas.

Both classes are well defined, because symbolic expressions for each separate term in a pair always coincide within the class but differ between different classes. The representatives of both classes were included in Table 3. In the first class only grade-3 and grade-4 cancellation can occur between both terms in a pair. In the second class, the MV in each pair generally is made up of grades 1, 4 and 5, which cancel out in the final result (see Example 4). It is also interesting to note that formulas of Table 4 can also be rewritten as a sum of three different terms with all weight coefficients being the same and equal to $1/3$ as shown in Table 5. The inverse MV formula can be constructed by taking any three expressions either from sets S_1, S_2 and S_3, or from sets T_1, T_2 and T_3 listed in Table 5. For example, the formulas

$$\frac{1}{3}H(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}} + \frac{1}{3}H((H_{\bar{4}}H_{\bar{4}})_{\bar{4}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}} + \frac{1}{3}HH(HH)_{\bar{1},\bar{4},\bar{5}} \quad \text{and}$$
$$\frac{1}{3}H(H_{\bar{4},\bar{5}}(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{4}})_{\bar{1},\bar{4}} + \frac{1}{3}H((H_{\bar{1},\bar{4}}H_{\bar{4},\bar{5}})_{\bar{4}}H_{\bar{4},\bar{5}})_{\bar{1},\bar{4}} + \frac{1}{3}H(H(HH_{\bar{1},\bar{5}})_{\bar{4}})_{\bar{1},\bar{5}}$$

are pretty valid choices.

Generally the Table 5 allows to make $4^3 = 64$ different triplets from sets S_i and $2^3 = 8$ triplets from sets T_i. Thus, all in all we can construct $64 + 8 = 72$ different triplet formulas for inverses in the case $n = 6$ without taking into account possible reversed forms.

Table 5. The sets for construction of different triplets of weight $\frac{1}{3}$ in inversion formulas for $n=6$

$S_1 =$	$\{\frac{1}{3}H(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}},\quad \frac{1}{3}H(H_{\bar{4}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{4}})_{\bar{4}},$ $\frac{1}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{4}}H_{\bar{4}})_{\bar{4}})_{\bar{1},\bar{4},\bar{5}},\quad \frac{1}{3}H(H_{\bar{1},\bar{4},\bar{5}}(H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}}\},$
$S_2 =$	$\{\frac{1}{3}H((H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}}H_{\bar{4}})_{\bar{4}},\quad \frac{1}{3}H((H_{\bar{4}}H_{\bar{4}})_{\bar{4}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}},$ $\frac{1}{3}H((H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{4}}H_{\bar{4}})_{\bar{4}},\quad \frac{1}{3}H((H_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}}H_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{4},\bar{5}}\},$
$S_3 =$	$\{\frac{1}{3}H(H_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{1},\bar{4},\bar{5}})_{\bar{1},\bar{5}},\quad \frac{1}{3}H(H_{\bar{1},\bar{5}}(HH)_{\bar{4}})_{\bar{1},\bar{5}},$ $\frac{1}{3}HH(H_{\bar{1},\bar{5}}H_{\bar{1},\bar{5}})_{\bar{4}}, \frac{1}{3}HH(HH)_{\bar{1},\bar{4},\bar{5}}\}.$
$T_1 =$	$\{\frac{1}{3}H(H_{\bar{1},\bar{4}}(H_{\bar{1},\bar{4}}H_{\bar{4},\bar{5}})_{\bar{4}})_{\bar{4},\bar{5}},\quad \frac{1}{3}H(H_{\bar{4},\bar{5}}(H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{4}})_{\bar{1},\bar{4}}\},$
$T_2 =$	$\{\frac{1}{3}H((H_{\bar{1},\bar{4}}H_{\bar{4},\bar{5}})_{\bar{4}}H_{\bar{4},\bar{5}})_{\bar{1},\bar{4}},\quad \frac{1}{3}H((H_{\bar{4},\bar{5}}H_{\bar{1},\bar{4}})_{\bar{4}}H_{\bar{1},\bar{4}})_{\bar{4},\bar{5}}\},$
$T_3 =$	$\{\frac{1}{3}H(H(HH_{\bar{1},\bar{5}})_{\bar{4}})_{\bar{1},\bar{5}},\quad \frac{1}{3}HH_{\bar{1},\bar{5}}(H_{\bar{1},\bar{5}}H)_{\bar{4}}\}.$

For details see the text

Example 3. In $Cl_{4,2}$ let's take $\mathsf{A} = 2 + \mathbf{e}_1 + \mathbf{e}_5 - 2\mathbf{e}_{15} + 3\mathbf{e}_{26} + 3\mathbf{e}_{1256}$ and find the inverse with formula from Table 3, $\mathsf{AG} = \frac{1}{3}HH(HH)_{\bar{1},\bar{4},\bar{5}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}}$, which has a minimal number of negations. We obtain $H = \mathsf{A}\mathsf{A}_{\bar{2},\bar{3},\bar{6}} = 8\mathbf{e}_1 + 8\mathbf{e}_5$. In the next step, however, we get zero since $HH = 0$ as well as $H_{\bar{4}}H_{\bar{4}} = 0$. In fact, the MV in the considered example was constructed multiplying the MV $1 + 2\mathbf{e}_1 + 3\mathbf{e}_{126}$ by isotropic vector $(\mathbf{e}_1 + \mathbf{e}_5)$ from the left.

It is easy to check that the last mentioned MV $\mathsf{A}' = 1 + 2\mathbf{e}_1 + 3\mathbf{e}_{126}$ is non-invertible as well. Indeed, $H = \mathsf{A}'\mathsf{A}'_{\bar{2},\bar{3},\bar{6}} = -4 + 4\mathbf{e}_1$, then $HH = 32 - 32\mathbf{e}_1$, and finally $HH(HH)_{\bar{1},\bar{4},\bar{5}} = 0$. In a similar way $H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}} = 0$. One can check that all formulas in Table 4 will yield the same result, namely, zero. In [9] one can find more information on MVs that contain isotropic multipliers.

Example 4. In $Cl_{1,5}$ let's find inverse of $\mathsf{A} = 2 + \mathbf{e}_1 + 4\mathbf{e}_3 + \mathbf{e}_{15} + 3\mathbf{e}_{126}$ using the same generic formula with minimal number of negations. Computation steps are:

1. $H = \mathsf{A}\mathsf{A}_{\bar{2},\bar{3},\bar{6}} = -3 + 4\mathbf{e}_1 + 16\mathbf{e}_3 - 2\mathbf{e}_5 - 24\mathbf{e}_{1236}$,
2. $HH = -811 - 24\mathbf{e}_1 - 96\mathbf{e}_3 + 12\mathbf{e}_5 + 144\mathbf{e}_{1236} - 96\mathbf{e}_{12356}$,
3. $\frac{1}{3}HH(HH)_{\bar{1},\bar{4},\bar{5}} = 1/3(678025 + 27648\mathbf{e}_5 + 2304\mathbf{e}_{1236} + 18432\mathbf{e}_{1256} - 4608\mathbf{e}_{2356} + 3456\mathbf{e}_{12356})$,
4. $(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}} = -811 + 24\mathbf{e}_1 + 96\mathbf{e}_3 - 12\mathbf{e}_5 + 144\mathbf{e}_{1236} - 96\mathbf{e}_{12356}$,
5. $(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}} = -2487 - 3316\mathbf{e}_1 - 13264\mathbf{e}_3 - 646\mathbf{e}_5 + 19704\mathbf{e}_{1236} - 1536\mathbf{e}_{1256} + 384\mathbf{e}_{2356} + 864\mathbf{e}_{12356}$,
6. $\frac{2}{3}H(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}} = 2/3(678025 - 13824\mathbf{e}_5 - 1152\mathbf{e}_{1236} - 9216\mathbf{e}_{1256} + 2304\mathbf{e}_{2356} - 1728\mathbf{e}_{12356})$.

From (3) and (5) we get that determinant norm of A is equal to 678025. Doing calculations in a similar way we find the inverse:
$\mathsf{A}^{-1} = \frac{1}{678025}\big(\frac{1}{3}(44766 - 9765\mathbf{e}_1 - 95588\mathbf{e}_3 + 1841\mathbf{e}_{15} + 8412\mathbf{e}_{26} - 5176\mathbf{e}_{35} - 71355\mathbf{e}_{126} - 12112\mathbf{e}_{135} + 20568\mathbf{e}_{236} - 1554\mathbf{e}_{1256} + 19608\mathbf{e}_{2356} - 7488\mathbf{e}_{12356}) +$

$\frac{2}{3}(44766 - 9765\mathbf{e}_1 - 95588\mathbf{e}_3 + 1841\mathbf{e}_{15} + 8412\mathbf{e}_{26} + 8\mathbf{e}_{35} - 71355\mathbf{e}_{126} - 12112\mathbf{e}_{135} + 18840\mathbf{e}_{236} - 8466\mathbf{e}_{1256} + 21336\mathbf{e}_{2356} - 4032\mathbf{e}_{12356})\big) = \frac{1}{678025}(44766 - 9765\mathbf{e}_1 - 95588\mathbf{e}_3 + 1841\mathbf{e}_{15} + 8412\mathbf{e}_{26} - 1720\mathbf{e}_{35} - 71355\mathbf{e}_{126} - 12112\mathbf{e}_{135} + 19416\mathbf{e}_{236} - 6162\mathbf{e}_{1256} + 20760\mathbf{e}_{2356} - 5184\mathbf{e}_{12356})$.

If a different formula from Table 4 is employed in finding the inverse, for example, using the first line $\frac{1}{3}H(H(HH_{\bar{1},\bar{5}})_{\bar{4}})_{\bar{1},\bar{5}} + \frac{2}{3}H(H_{\bar{1},\bar{4}}(H_{\bar{1},\bar{4}}H_{\bar{4},\bar{5}})_{\bar{4}})_{\bar{4},\bar{5}}$, then intermediate results will be different. In particular, for the first term $\frac{1}{3}H(H(HH_{\bar{1},\bar{5}})_{\bar{4}})_{\bar{1},\bar{5}}$ we find 678025/3, and for the second term $\frac{2}{3}H(H_{\bar{1},\bar{4}}(H_{\bar{1},\bar{4}} H_{\bar{4},\bar{5}})_{\bar{4}})_{\bar{4},\bar{5}}$ we find 1356050/3. Thus, in this case no cancellation between nonzero grades of both terms in the pair occurs.

Example 5. This example shows that, in principle, one can use the formula valid for $n = 6$ to find the determinant norms and inverses of MVs in all smaller algebras (for $n < 6$) as well. At first sight this may be a bit unexpected. The following example explains how it happens. Let us take the MV $\mathsf{A} = 45 + 55\mathbf{e}_1 + 84\mathbf{e}_{12} + 39\mathbf{e}_{134} + 93\mathbf{e}_{234} + 15\mathbf{e}_{1234}$ in $Cl_{2,2}$ and use the determinant norm formula for $n = 6$ instead of $n = 4$: $\mathsf{AG} = \frac{1}{3}HH(HH)_{\bar{1},\bar{4},\bar{5}} + \frac{2}{3}H(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}}$, where $H = \mathsf{AA}_{\bar{2},\bar{3},\bar{6}}$. The negation of absent grades, of course, should be ignored.

Computation steps are:

1. $H = 22501 + 7740\mathbf{e}_1 - 10410\mathbf{e}_2 - 8880\mathbf{e}_{1234}$,
2. $HH = 753425101 + 348315480\mathbf{e}_1 - 468470820\mathbf{e}_2 - 399617760\mathbf{e}_{1234}$,
3. $\frac{1}{3}HH(HH)_{\bar{1},\bar{4},\bar{5}} = 67166445910339801/3$,
4. $H_{\bar{4}}H_{\bar{4}} = 753425101 + 348315480\mathbf{e}_1 - 468470820\mathbf{e}_2 + 399617760\mathbf{e}_{1234}$,
5. $\frac{2}{3}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}} = 2 * 67166445910339801/3$,
6. $\mathsf{AG} = 67166445910339801 = (259164901)^2$,

where $\sqrt{\mathsf{AG}} = 259164901$ is exactly the determinant norm of the MV calculated by the formula valid for $n = 4$. So, the calculation using the norm formula for $n = 6$ yields the squared determinant of a matrix representation of the multivector in $Cl_{2,2}$. It is easy then to check that the formula valid for $n = 6$ gives the inverse of this multivector as well. Indeed,

7) $H(HH)_{\bar{1},\bar{4},\bar{5}} + 2(H_{\bar{4}}(H_{\bar{4}}H_{\bar{4}})_{\bar{1},\bar{4},\bar{5}})_{\bar{4}} = 17494408312203 - 6017809001220\mathbf{e}_1 + 8093719858230\mathbf{e}_2 + 6904152962640\mathbf{e}_{1234}$
8) Finally, in the case $n = 6$, the inverse MV calculated in $Cl_{2,2}$ is

$$\begin{aligned}
\mathsf{A}^{-1} =& \frac{1}{3 * 67166445910339801}\big(559831173421635 - 630567641500575\mathbf{e}_1 \\
&+ 127983403060830\mathbf{e}_2 - 1024375706022402\mathbf{e}_{12} + 61927453093950\mathbf{e}_{34} \\
&- 560876126302467\mathbf{e}_{134} - 1156984425071379\mathbf{e}_{234} \\
&- 302208303582585\mathbf{e}_{1234}\big) \\
=& \frac{1}{259164901}\big(720045 - 811025\mathbf{e}_1 + 164610\mathbf{e}_2 - 1317534\mathbf{e}_{12} + 79650\mathbf{e}_{34} \\
&- 721389\mathbf{e}_{134} - 1488093\mathbf{e}_{234} - 388695\mathbf{e}_{1234}\big)
\end{aligned}$$

which does coincide with the inverse MV calculated in $Cl_{2,2}$ by means of the inversion formula for $n = 4$.

And this is not a coincidence. Explicit symbolic computations confirm that the determinant norm formula for $n = 6$, when applied to a MV in the algebra of a space of dimension $n < 6$, yields either the determinant of a matrix representation of MV (for $n = 5$), or the determinant raised to the power 2 in the case of the dimension $n = 3$ or $n = 4$, or to the power 4 for $n = 2$ and $n = 1$. Exactly the same happens, when the determinant norm formula for $n = 5$ is applied to the cases $n = 4, 3, 2$ and to the case $n = 1$. Similarly, when the determinant norm formula for $n = 4$ is used in the case $n = 3$, we obtain the determinant of a matrix representation of the MV, whereas the same procedure for $n = 2$ and $n = 1$ yields the determinant raised to the power 2, etc. So, the determinant norms disclose very interesting onion-like structure, where each higher dimensional formula embraces all lower dimensional ones. We think that this property may be important in further investigations.

6. Conclusions

From this computer-aided study we conjecture that the inverse of a general MV in a Clifford algebra can be always expressed as a linear combination of some number of suitably constructed self-negated products, where the initial MV must stay in the left-most or right-most position. For the first time we have explicitly derived compact formulas for inverse MVs that consist of sums of two grade-negated products (Tables 3, 4) for all Clifford algebras of spaces of dimension $n = 6$. The formulas are independent of the signature (p, q) and may be useful in numerical and especially in symbolic programming. They reduce to known expressions of inverses for vectors, bivectors, etc, or their combinations.

We have also presented different formulas for inverses in even Clifford subalgebras for dimensions $n \leq 6$ (Table 2), which are important for spinor algebras. A number of previously unknown explicit expressions were provided for algebras of spaces of dimension $n = 5$ (Table 1), some of which may be interesting from computational point of view as well. We believe, the formulas in Table 1 are all possible formulas that give the determinant with a single term in case $n = 5$ by using only geometric products and involutions. We have shown that determinant norm formulas for $n = 6$ split into two classes, Tables 4 and 5. We have also presented inverses in the case $n = 6$ in a more symmetric linear combination of three terms with weights all equal to $1/3$ (Table 5). Though this form is not preferred from computational point of view it may be interesting when searching similar formulas for higher dimension Clifford algebras. The computation of inverse MVs by the formulas in Table 3 requires a considerably smaller number of multiplications since many pieces in the formulas are iterated many times. The speed of calculation can be further improved if the MV multiplication algorithm is realized in such a way, that it can skip calculation of some specific grades, because we know in advance that some of the grades should vanish from the product. In this case a large number of multiplications can be avoided.

When the determinant norm is expressed as a sum of self-negated products, the minimal number of factors in each product is equal to the order of the matrices that represent the multivectors according to the 8-periodicity table [11]. In the case $n = 2$, the determinant norm consists just of two multipliers, whereas for $n = 3$ as well as for $n = 4$ one has to make products of four MV multipliers in accordance with the order of the corresponding matrices in the 8-periodicity table. The Clifford algebras of dimension 5 or 6 require products of eight factors to construct the determinant norm. Then, looking at the 8-periodicity table, one may expect that the determinant norm of algebras of spaces of dimension 7 or 8 can be constructed by linear combinations of products of 16 factors, etc.

From the considerations above it should be clear that the complexity of finding an explicit expression of the inverse of a general MV grows rapidly with the dimension n. For example, when $n = 6$ a direct head-on symbolic multiplication of eight general arbitrary multivectors in the worst case requires $(2^6)^8 = 281\,474\,976\,710\,656$ geometric multiplications of basis elements, which fortunately can be bypassed due to significant grade reduction in the determinant formula as discussed in this article. However, in practice it does not help much when searching for the correct shape of the determinant norm formula. This is why it is worth first trying reduced (shorter) MVs which, under good choice, may greatly reduce the number of candidates for possible linear combinations. It should be stressed that the determinant norms and inverse multivectors obtained in this paper make computations self-contained within the geometric product and negation structure without any need to resort to matrix representations.

Finally, we have demonstrated that the formula for $n = 6$ alone produces determinant norms and inverses of MVs in all algebras of spaces of dimension $n \leq 6$ and of any signature, *albeit* using more geometric multiplications. The identified onion-like (i.e. telescoping) structure of determinant norms and the numerous shapes of inverse multivectors presented in this article incline us to believe that similar formulas (which include only operations of grade negation, geometric product and summation) may exist for Clifford algebras of all dimensions.

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